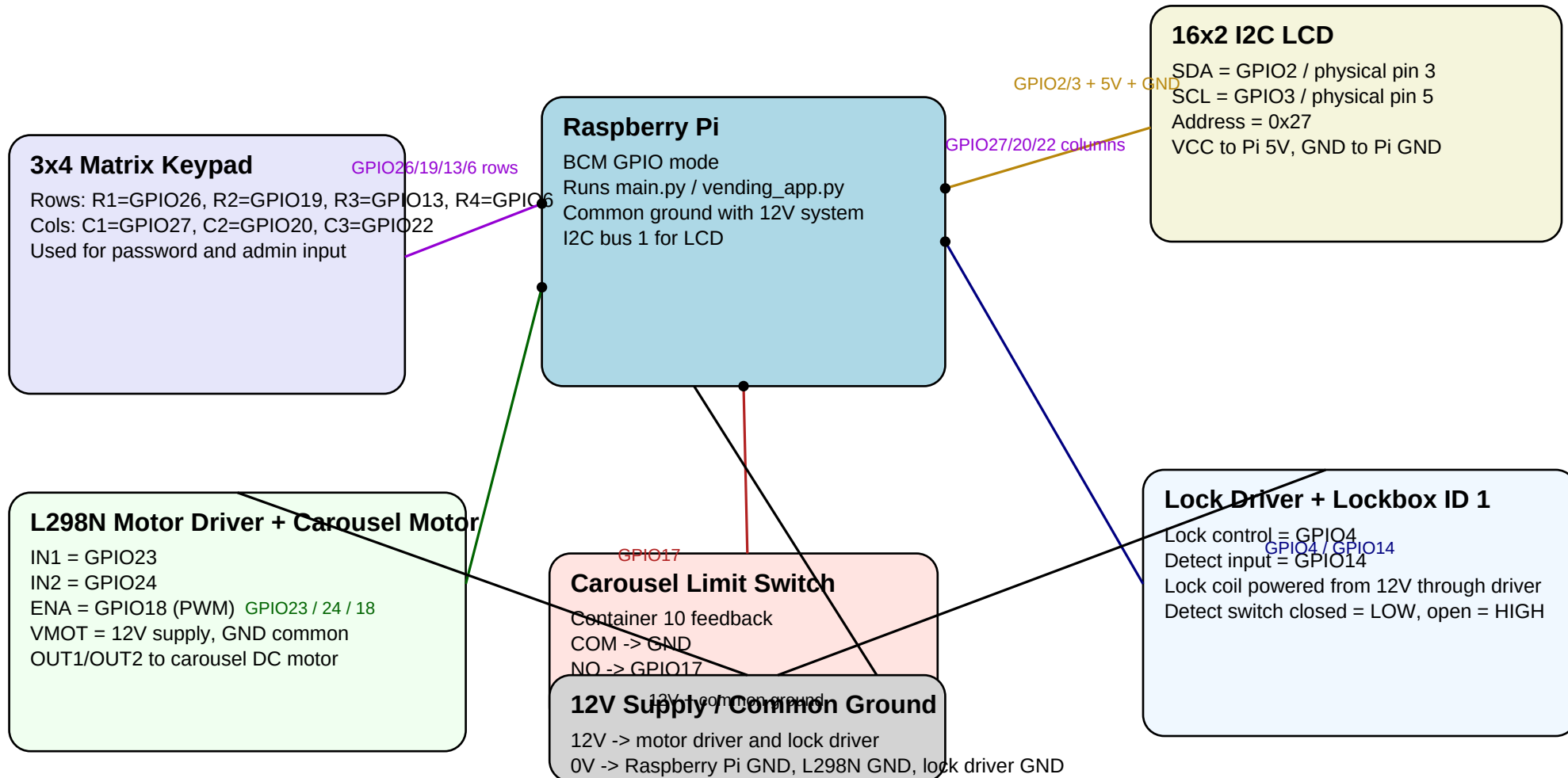


ECE455 Current Wiring Schematic

Based on the current connected devices in /home/pi/ECE455

Active physical devices: lockbox ID 1 and carousel ID 10



Note: only lockbox ID 1 and carousel ID 10 are currently installed/controlled in this build.

Connection Table and Build Notes

Device	Signal / Wire	BCM	Physical Pin	Destination / Notes
Keypad	R1	26	37	Matrix keypad row 1
Keypad	R2	19	35	Matrix keypad row 2
Keypad	R3	13	33	Matrix keypad row 3
Keypad	R4	6	31	Matrix keypad row 4
Keypad	C1	27	13	Matrix keypad column 1
Keypad	C2	20	38	Matrix keypad column 2
Keypad	C3	22	15	Matrix keypad column 3
LCD	SDA	2	3	I2C bus 1
LCD	SCL	3	5	I2C bus 1
LCD	VCC	5V	2/4	Pi 5V rail
LCD	GND	GND	6/9/14/etc.	Pi ground
Motor driver	IN1	23	16	L298N input
Motor driver	IN2	24	18	L298N input
Motor driver	ENA	18	12	L298N PWM enable
Motor driver	VMOT	12V	--	External 12V supply
Limit switch	NO	17	11	Carousel container 10 input, pull-up enabled
Limit switch	COM	GND	--	Switch common to ground
Lockbox 1	Lock control	4	7	Driver input to 12V lock path
Lockbox 1	Detect	14	8	Closed=LOW, open=HIGH
System	Common ground	--	--	Pi, L298N, and lock driver ground tied together

Build notes

1. Keep all GPIO wiring as signal-level only; do not power motors or locks from Pi GPIO.
2. Use a shared ground between the Pi, L298N, and lock driver.
3. The project currently controls only lockbox ID 1 and carousel ID 10.
4. If the wiring changes, update config_vm.py and regenerate this PDF.